STUDENT INTENSIVE

Aryabhat Quiz: Preliminary Preparation

Fourteen weekly classroom sessions for classes 6–10, walking the quiz's own study material end to end — every session mapped to the PDFs the written preliminary round is set from.

14	91	26 h 30 min	Amateur Astronomy Club
SESSIONS	CONCEPTS TAUGHT	TOTAL	AUDIENCE

ASSUMED ALREADY KNOWN

Day & Night · Seasons · Phases of the Moon · Observing the Moon · Light & Shadow · Earth is a Sphere · Spin & Axis · The Pole Star · Stars Differ · Diurnal Motion · The Heliocentric Model · Planets: the Wanderers · The Solar System Family · The Sun Powers Earth · Latitude & Longitude · Maps & Scale Models · Angles & the Protractor · Big Numbers & Powers of Ten · Models in Science · Gravity: Things Fall · Free Fall & the Value of g · Newton's Second Law · Conservation of Energy · Speed, Distance & Time · Spectrum & Colour · Lenses, Mirrors & Focal Length · Hot & Cold: Temperature · Conduction, Convection & Radiation · Waves · The Ocean of Air · The Nucleus: Protons, Neutrons & Isotopes

Closed under prerequisites: 56 concepts in total are taken as prior knowledge.

1 Week 1 — The Sun, our reference star

CLASSROOM · 2 H

KIT Projector · Paper, pencils, protractor

L2 Light & Lookback Time

Light is fast but finite — 8 minutes from the Sun, 4 years from the nearest star. Every telescope is a time machine: we see objects as they were, not as they are.

L2 The Sun is a Star

Our Sun is an ordinary star seen close-up — a million-km fusion furnace. Every night-sky star is a sun; many have their own planets.

L3 Anatomy of the Sun

Outwards from a 15-million-kelvin fusion core: a radiation zone, a boiling convection zone, then the photosphere we call the surface — mottled like a pot of cooking oatmeal — the thin chromosphere, and the corona. The corona is the puzzle: at millions of degrees it is hundreds of times hotter than the surface beneath it, and nobody has fully explained why.

L3 How Stars Shine

Stars fuse hydrogen into helium; the missing mass becomes energy by $E = mc^2$. The Sun converts four million tonnes of matter to light every second — and has fuel for five billion years more.

L3 Sunspots & Solar Activity

The Sun's magnetic weather: spots, flares and mass ejections rising and falling on an 11-year cycle. It drives aurorae and can rattle satellites and power grids.

L4 Beyond Visible Light

Radio, infrared, ultraviolet, X-ray, gamma — visible light is one octave of a vast keyboard. Each band needs its own telescope and reveals a different universe.

L3 Solar Wind & the Heliosphere

A thin stream of charged particles is driven off the Sun's surface by light pressure, electric charge and magnetic flux, and can be detected right across the solar system. The bubble it inflates against interstellar space — the heliosphere — is the real edge of the Sun's domain.

L3 Space Weather

When the Sun belches a coronal mass ejection at Earth, the result is a geomagnetic storm: power lines damaged on the ground, long-range communications disrupted, astronauts retreating to the deepest part of their spacecraft — and aurorae bright enough to make it all look worth it.

TRAINER NOTE Study_Material_1 opens at 'main sequence G2 dwarf' with no ramp — this session IS the ramp. Drill the layer order (core → radiative zone → convective zone → photosphere → chromosphere → corona) both directions; the quiz loves ordering questions. Numbers to own: 5800 K photosphere, the 11-year sunspot cycle, 8 minutes for sunlight to reach us — which is why light-travel is taught here, not in the stars weeks. The corona shows up in X-ray images and the chromosphere in H-alpha, which is all the excuse needed to lay out the electromagnetic spectrum once, properly; three later sessions stand on it. Keep flares and CMEs distinct — one is light, one is stuff — and hang the aurora off the solar wind while you are there.

2 Week 2 — The celestial sphere and the constellations

INDOOR LAB · 2 H

KIT Star chart / planisphere · Projector · Paper, pencils, protractor

L2 Angles on the Sky

Sky positions and sizes are measured in angles: your fist at arm's length spans about 10°, the Moon half a degree. Degrees, arcminutes and arcseconds are the astronomer's ruler.

L1 Constellations

Star patterns used as sky landmarks — 88 official constellations today, with older patterns in every culture. They are direction labels, not physical groups.

L2 The Milky Way Band

From a dark site a faint luminous band arches across the sky — the combined light of billions of stars in our galaxy's disc, seen from inside it.

L1 Asterisms

Popular star patterns that aren't official constellations — the Big Dipper is part of Ursa Major, the Summer Triangle spans three constellations. The sky's informal landmarks.

L2 Star Names & Bayer Letters

Bright stars carry ancient proper names (many Arabic: Aldebaran, Deneb); astronomers label stars α, β, γ by brightness within each constellation. Two naming systems, one sky.

L2 Star Charts & Planetarium Apps

Reading a sky map: orientation, scale, magnitudes, and matching chart to sky. A planisphere or app turns the catalogue sky into 'what's above me right now'.

L3 Celestial Sphere

A model sphere of infinite radius carrying the stars, with celestial poles and a celestial equator projected from Earth's own. It turns star positions into geometry we can compute.

L3 Sky from Different Latitudes

At the equator every star rises and sets vertically; at the poles nothing rises or sets at all. Your latitude completely decides which sky you own — Canopus for Chennai, none for London.

TRAINER NOTE Charts on every desk — Study_Material_2's opening, pulled forward because the ecliptic next week stands on it. Fist at arm's length = 10°, then make them measure chart distances with it. The plants for the quiz: 88 constellations fixed by the IAU in 1930 (memorise the number and the date); the Big Dipper is an ASTERISM, not a constellation — the classic trap; Bayer's alpha-beta lettering on the charts in front of them; Canopus never rises above 37°N, and circumpolar stars never set — both are the same latitude argument, run it once each way. The Milky Way band gets ten honest minutes: most of the room has never seen it, and Week 11 builds the Galaxy out of it.

3 Week 3 — The ecliptic, the nodes, and why eclipses skip months

INDOOR LAB · 2 H

KIT Projector · Paper, pencils, protractor

L2 Moon Orbits Earth

The Moon circles Earth roughly monthly at about 384,000 km — around 30 Earth-diameters away. Its orbital motion is what drives the phase cycle and its nightly 13° eastward drift.

L2 Seasonal Skies

The night sky changes with the calendar — Orion rules winter evenings, Scorpius the monsoon sky — because Earth's orbit points our nights toward different stars each season.

L2 Lunar Eclipses

At full moon, the Moon can pass through Earth's shadow and turn copper-red — sunset light bent through our atmosphere. Safe to watch, visible from the whole night side.

L2 Solar Eclipses

At new moon, the Moon's small shadow can sweep across Earth, briefly hiding the Sun. Totality happens because the Moon and Sun span the same half-degree — a cosmic coincidence.

L2 Ecliptic & Zodiac

The Sun's yearly path against the stars is the ecliptic; the belt of constellations it crosses is the zodiac. The Moon and planets always keep near this line.

L4 Precession of the Equinoxes

Earth's axis slowly cones like a spinning top, once in about 25,800 years. Pole stars change (Thuban → Polaris → Vega) and the equinox point slides westward through the zodiac.

L4 Eclipse Seasons & Nodes

The Moon's orbit is tilted 5°, so eclipses happen only when full/new moon falls near the orbit's crossing points (nodes) — twice a year. The 18-year Saros cycle repeats the pattern.

L4 Rahu & Ketu: the Nodes

The 'shadow planets' of Indian astronomy are the Moon's ascending and descending nodes — precisely where eclipses occur. Myth and orbital mechanics describing the same two points.

TRAINER NOTE One argument, start to finish: the constellations parade with the seasons because the Sun walks the ecliptic; the Moon's orbit is tilted 5.1° to that line; so shadows usually miss, and only at the nodes do they land. Two hoops tilted 5.1° apart settle in one minute what no diagram settles in ten. Rahu and Ketu ARE the two nodes — the material and the quiz both pair the Indian framing with the geometry, so teach them as one thing. The 400×/400× size coincidence is angular measure from last week earning its keep. Precession rides along on the zodiac: Vega was the pole star 12,000 years ago and will be again — a certain quiz fact. Kill the classic confusion out loud: phases are geometry we see every month, eclipses are shadows we mostly don't.

4 Week 4 — What is a planet?

CLASSROOM · 1 H 30 MIN

KIT Projector · Paper, pencils, protractor

L3 Scale of the Solar System

With Earth as a peppercorn, the Sun is a football 25 m away and Neptune is 750 m down the road. The emptiness is the real lesson; the AU is the ruler.

L3 Terrestrial & Jovian Planets

The inner four share a build — dense core, rocky mantle, thin crust — because the newly ignited Sun lashed them with a titanic solar wind that stripped their hydrogen and helium away. Farther out the wind abated, the growing planetesimals kept their gas, and hoarded so much hydrogen that it compressed into metallic forms.

L3 What Counts as a Planet

In 2006 more than 2500 astronomers at the IAU's 26th General Assembly passed Resolution 5A and set three tests: it must move in a definite orbit around the Sun, its own gravity must have pulled it into a nearly round hydrostatic shape, and it must have cleared the neighbourhood of its orbit. Pluto passes the first two — which is how nine planets became eight.

L3 Dwarf Planets

A dwarf planet passes every test but one: round, orbiting the Sun, not anybody's satellite — but it has not cleared its orbital neighbourhood. Ceres works the asteroid belt at 413 million km; Pluto, Haumea, Makemake and Eris lie out beyond Neptune, Eris the farthest at some 14,410 million km.

TRAINER NOTE The material quotes IAU Resolution 5A's three tests verbatim and so will the quiz — have them recite it: orbits the Sun, round under its own gravity, has cleared its neighbourhood. Pluto fails the third; that is the whole story, tell it without the outrage. The planet statistics table is pure memorisation — set it as homework and quiz it cold next week. Terrestrial versus jovian is the reasoning half, and scale is the shock half: shrink the Sun to a football and walk the classroom to Neptune. Saturn's rings get their sentence here (all four giants have rings; Saturn's are just the ones you can see) — the quiz asks 'which planets have rings' and accepts only 'all four'.

5 Week 5 — Orbits and gravity

CLASSROOM · 2 H

KIT Projector · Paper, pencils, protractor

L3 **Newton's Third Law**

Forces always come in pairs: if you push the wall, the wall pushes you, equally and oppositely. The pair acts on DIFFERENT bodies — which is exactly why they never cancel each other out.

L4 **Newton's Law of Gravitation**

Every mass attracts every other with a force $\propto m_1m_2/r^2$, and the same constant G that makes an apple drop holds the Moon in orbit. Two lumps of matter anywhere in the universe are already pulling on each other.

L3 **Planetary Configurations**

Opposition, conjunction, greatest elongation: the geometry words that say where a planet is relative to the Sun — and therefore when it rises and how bright it looks.

L4 **Kepler's Laws**

Orbits are ellipses with the Sun at a focus; planets sweep equal areas in equal times; period squared tracks distance cubed. Three rules that turned astronomy quantitative.

L4 **Escape Velocity**

Throw something upward fast enough and it never comes back. Set the kinetic energy $\frac{1}{2}mv^2$ against the gravitational potential energy GMm/R and the break-even speed drops out — 11.2 km/s for Earth, 4.25 for Mercury, 59.6 for Jupiter. It depends entirely on the planet and not at all on what you threw.

L4 **Universal Gravitation**

Newton showed one inverse-square force explains falling apples, the Moon's orbit and Kepler's laws together. Gravity is the engine of everything astronomy studies.

L4 **How the Solar System Formed**

Roughly 4.5 billion years ago a disk of gas and dust spun around the proto-Sun; whirls in it grew into planetesimals, heavy elements sank to their centres, and the large ones swept up the small until only big bodies on near-circular orbits were left. The crossfire it took to settle is still written on every old surface as craters.

TRAINER NOTE Kepler qualitatively — ellipse, equal areas, farther-is-slower — then Newton as the why, and the third law banked deliberately: Week 13's rockets are nothing else. Escape velocity is the planet table's most quiz-mined column (11.2 km/s for Earth) and the road to black holes in Week 10 — say so now. Opposition, conjunction, elongation: the material covers the configurations but NEVER explains retrograde looping — a known hole in the PDFs, close it with three students walking concentric circles. Formation closes the loop on last week: the frost line is WHY the terrestrial/jovian table looks the way it does. One line for Mercury's precessing perihelion needing Einstein; the material name-drops it.

6 Week 6 — The Moon, tides, and everyone else's moons

CLASSROOM · 2 H

KIT Projector · Paper, pencils, protractor

L2 Craters & Maria

The Moon's dark 'seas' are ancient lava plains; its craters are asteroid scars, preserved for billions of years with no air or water to erase them. The face of the Moon is a history book.

L3 How the Moon Formed

Only one story survives rigorous modelling: early Earth took a glancing blow from a proto-planet about the size of Mars, and the debris — largely Earth's own mantle — gathered into a dense ring and coalesced. The Moon itself is the evidence: mantle rock much like ours, but moonquake seismometry and the missing magnetic field both say there is no metal core.

L3 Tidal Locking

The Moon rotates exactly once per orbit, so one face is forever turned to Earth. Tidal friction braked it into this lock — the fate of most large moons.

L3 Tides

The Moon pulls Earth's near side harder than its far side, raising two water bulges — hence two tides a day. Sun-Moon alignment makes spring tides; right angles make neaps.

L3 Galilean Moons

Jupiter's four big moons shuffle visibly from night to night — a miniature solar system in any small telescope, and Galileo's proof that not everything orbits Earth.

L3 Moons of the Solar System

The tally is lopsided and still climbing: as on 30 June 2026, Saturn 285, Jupiter 115, Uranus 28, Neptune 16, Mars 2, Earth 1, and nothing at all for Mercury or Venus. Titan is the only satellite with a substantial atmosphere; Europa's cracked ice appears to float on a worldwide ocean.

TRAINER NOTE Maria, highlands and craters first — you cannot tell the giant-impact story to someone who has never looked at what it built. Tides as the material argues them: an imbalance across the Earth's width, hence TWO bulges, hence two tides a day — the commonest wrong answer in this section is 'the water is pulled toward the Moon, so one bulge'. The ledger closes it: Earth's spin slows, the Moon recedes 3.8 cm a year, angular momentum balances the books. Tidal locking explains the far side — which is NOT a dark side, say it. The Galilean four are memorisation (order from Jupiter: Io, Europa, Ganymede, Callisto), with Io–Europa–Ganymede's 4:2:1 clockwork as the showpiece fact.

7 Week 7 — Small bodies: asteroids, comets, meteors

CLASSROOM · 1 H 30 MIN

KIT Projector · Paper, pencils, protractor

L2 Asteroids & the Kuiper Belt

Rocky leftovers between Mars and Jupiter, icy ones beyond Neptune (Pluto's realm). Small-body surveys are one field where amateurs still make discoveries.

L2 Comets

Dirty snowballs on long elliptical orbits that grow tails — always pointing away from the Sun — when sunlight boils their ices. Halley returns every 76 years.

L2 Meteors & Meteor Showers

Shooting stars are dust grains burning up 80–100 km overhead. When Earth crosses a comet's debris trail we get an annual shower, radiating from one point among the stars.

L3 Impacts on Earth

Earth gets hit too — Lonar Lake in Maharashtra is a 50,000-year-old impact crater, and the dinosaurs met a 10-km asteroid. Erosion, not luck, hides our scars.

L4 The Oort Cloud

The outermost layer of the solar system is a spherical shell of cometary nuclei about 50,000 AU out — some 8/10 of a light-year, far enough that the Sun's grip is feeble and a passing star can tug comets loose. It is the source of the long-period comets; the Kuiper Belt, by contrast, is a flat disc just outside Neptune.

L3 Near-Earth Objects & Planetary Defence

Some asteroids cross Earth's orbit — Toutatis came within 6 Moon distances in 2004 — and the one that finished the dinosaurs was only mountain-sized, not planet-sized. That is the whole case for surveying: unlike a long-period comet, which we may not see until it is too late, an Earth-crossing asteroid can be catalogued decades early and nudged.

TRAINER NOTE The terminology trap is the guaranteed question: meteoroid in space, meteor in the air, meteorite on the ground — drill it until it bores them. Comet tails point AWAY from the Sun, not behind the motion; the quiz has asked. Geography to place, in order outward: main belt, Kuiper belt, Oort cloud. The material's proper nouns are fair game — Toutatis at six lunar distances, Shoemaker-Levy 9 into Jupiter, the Leonids from Tempel-Tuttle and what a radiant is, Vesta and Flora as named families. Craters tie last week's lunar face to this week's impactors — same story, two witnesses.

8 Week 8 — Magnitudes, parallax, and the ten brightest stars

INDOOR LAB · 2 H

KIT Projector · Paper, pencils, protractor

L3 **The Inverse-Square Law**

Double the distance, quarter the light — brightness falls as $1/d^2$. This single rule connects a star's true power, its distance, and how bright it looks to us.

L3 **The Magnitude Scale**

Astronomy's upside-down brightness scale: smaller numbers are brighter, and 5 magnitudes means 100×. Vega ≈ 0, faintest naked-eye stars ≈ +6, the Sun −26.7.

L3 **Star Colours & Temperature**

A star's colour is a thermometer reading: red ~3,000 K, yellow ~6,000 K, blue-white 10,000 K+. Hotter objects glow bluer — physics you can check with your own eyes.

L4 **Spectroscopy**

Elements imprint fingerprint lines on a spectrum. From those lines alone we read a star's composition, temperature and motion — how we know what stars are made of.

L4 **Spectral Classes: OBAFGKM**

Stars sort into temperature classes O (blue, 40,000 K) through M (red, 3,000 K); the Sun is a G2. One letter compresses a star's spectrum, colour and temperature.

L3 **Light Years & Cosmic Scale**

The nearest star is 4.2 light years away — 7,000× farther than Neptune. Grasping the jump from solar-system scale to interstellar scale is a rite of passage.

L4 **Stellar Parallax**

Nearby stars shift minutely against the background as Earth orbits — the parallax that defines the parsec and anchors every distance in astronomy.

L4 **Luminosity & Absolute Magnitude**

Apparent brightness mixes true output with distance; absolute magnitude removes distance (all stars imagined at 32.6 ly). Only then can stars be honestly compared.

TRAINER NOTE Magnitude arithmetic on paper until it is reflex: Hipparchus's scale runs BACKWARDS (lower is brighter, negatives brightest of all), 5 magnitudes = 100×. The Sun's pair is a certain question: −26.7 apparent, +4.8 absolute at 10 parsecs — and absolute magnitude only makes sense once the inverse-square law and parallax are on the table, so they are taught here, not assumed. The ten-brightest table is this course's biggest memorisation ask — name, constellation, magnitude, distance, colour — hand it out as a fortnight's drill, not a night's. The stories carry it: Bessel weighing Sirius by its wobble, Sirius B the first white dwarf, Betelgeuse lettered Alpha though Rigel outshines it — because it varies. OBAFGKM with the mnemonic of their choosing, and the Sun filed as G2 closes the circle to Week 1.

9 Week 9 — How stars live

CLASSROOM · 2 H

KIT Projector · Paper, pencils, protractor

L3 Nebulae & Star Clusters

Glowing gas clouds where stars are born; open clusters of young siblings; ancient globular swarms of hundreds of thousands. The Milky Way's visible ecology.

L5 Black-Body Radiation

Everything above absolute zero glows, and the glow reports its temperature twice over: total power per square metre climbs as T^4, while the peak wavelength slides shorter as it heats. Hotter is both brighter AND bluer — so red means cool, and a red giant is only luminous because luminosity is brightness-per-area multiplied by an area the size of Mars' orbit.

L4 Star Birth

Cold gas clouds collapse under gravity, heat up, and ignite fusion. The Orion Nebula is a stellar nursery in action, visible in binoculars.

L4 The H-R Diagram

Plot luminosity against temperature and stars fall into families: the main sequence, giants, white dwarfs. One scatter plot that encodes how stars live and die.

L4 Lives of Stars

Stars balance gravity against fusion for millions to trillions of years, then swell to giants and die — gently as white dwarfs or violently as supernovae. Mass decides everything.

L4 Mass–Luminosity Relation

A main-sequence star's luminosity goes as the fourth power of its mass: three times the mass, eighty-one times the light — and eighty-one times the rate of burning fuel. One exponent explains why heavy stars are both dazzling and doomed.

L4 Stellar Lifetimes

Time on the main sequence falls as the inverse cube of mass — fuel goes as M but burning goes as M^4, leaving M^{-3}. A 100-solar-mass star lasts about 10,000 years, the Sun 10 billion, a 0.08-solar-mass red dwarf 19.5 trillion.

TRAINER NOTE Study_Material_3's best chapter, and it reasons rather than lists — teach it that way. $L \propto M^4$ and $t \propto M^{-3}$: don't hand the exponents over, walk the argument (more mass → more weight to hold → hotter core → furiously faster burning; the material's hydrostatic-balance picture in one breath) so 'massive stars die YOUNG' lands as an inference — the quiz sets exactly that inversion as a trap. The red-giant paradox is the showpiece: cool AND bright is impossible for a point, so a red giant must be enormous — black-body reasoning doing real work. Plot Week 8's ten brightest on the HR diagram by hand. Birth bookends it: nebulae and clusters as the nurseries, and below the fusion floor, the brown dwarfs — 'failed star' is a phrase the quiz likes.

10 Week 10 — How stars die

CLASSROOM · 2 H

KIT Projector · Paper, pencils, protractor

L4 Planetary Nebulae & White Dwarfs

A dying Sun-like star exhales its outer layers into a glowing shell around a white-hot ember. The Ring Nebula is the Sun's own future, visible in a small telescope.

L4 Supernovae, Neutron Stars & Black Holes

Massive stars end in explosions that briefly outshine galaxies, forging heavy elements and leaving city-sized neutron stars or black holes. Your iron came from one.

L4 Black Holes

Squeeze a core past about 3 solar masses and gravity wins outright: escape velocity exceeds light speed, so nothing gets out past the event horizon. Not a cosmic vacuum cleaner — at a given distance its pull is exactly the pull that mass always had.

L4 Pulsars

A neutron star about 10 miles across, spinning hundreds to thousands of times a second with its surface near light speed, beams charged particles from its poles. If the beam sweeps our way we see clockwork flashes — regular enough that the first detection was half-seriously credited to aliens.

L4 Where the Elements Came From

Ordinary stars build elements only up to iron and return them on stellar winds and shed shells; everything heavier needs a supernova. The iron in your blood and the calcium in your bones were forged in a star that died — and the same debris is what makes rocky planets possible at all.

L5 The Chandrasekhar Limit

1.4 solar masses is the most a leftover core can weigh and still be propped up by electron degeneracy pressure as an Earth-sized white dwarf. Push past it and the electrons are forced into the protons to make neutrons; past about 3 solar masses, nothing holds at all.

L5 Type Ia & Type II Supernovae

Two very different ways to blow up a star: a white dwarf accreting past the Chandrasekhar limit (Type Ia), or a massive star's core collapsing when the fuel runs out (Type II). Both can outshine an entire galaxy for weeks and forge every element past the light ones — and a Ia's uniform brightness makes it a rung on the distance ladder.

TRAINER NOTE One flowchart, sorted by mass, and every ending hangs off it. The two thresholds are the quiz's favourite numbers in this section: 1.4 solar masses (Chandrasekhar — an Indian name on an Indian quiz, expect it) and ~3 for the neutron-star ceiling. A planetary nebula has nothing to do with planets — say it before a student says it wrongly in the hall. Nova versus supernova are genuinely different events and the material is explicit: a surface flash on a white dwarf versus the star destroyed; Type Ia versus Type II by mechanism. The material's marble-Earth argument kills the vacuum-cleaner misconception: collapse the Earth to a marble and the Moon's orbit doesn't change — a black hole pulls no harder than the star it was. End on nucleosynthesis: every atom past helium was cooked in a star — true, testable, and the line they will remember.

11 Week 11 — Deep sky: the Galaxy, other galaxies, and Messier's list

CLASSROOM · 2 H

KIT Projector · Paper, pencils, protractor

L3 Types of Nebula

Four kinds of cloud: emission, gas fluorescing under ultraviolet from hot young stars (M42 — the physics of a neon lamp); dark, dust blotting out what lies behind (the Horsehead, the Coal Sack); reflection, dust scattering a nearby star's light; and planetary, a shell shed by a dying low-mass star. The Horsehead manages all of them at once.

L3 Messier & NGC Catalogues

Messier's 110 'not-comets' are the sky's greatest-hits list; the NGC adds 7,840 more. Catalogue numbers are the shared language of deep-sky observing.

L4 Structure of the Milky Way

A barred spiral of 100–400 billion stars, 100,000 ly across, with us in a quiet arm 27,000 ly out. The naked-eye band is our inside view of the disc.

L4 Galaxies

Islands of billions of stars — spirals, ellipticals, irregulars — separated by millions of light years. Andromeda, our giant neighbour, is faintly visible to the naked eye.

L4 Supermassive Black Holes

Millions to billions of solar masses packed into a galaxy's heart — one is thought to sit at the centre of every galaxy. Ours, Sagittarius A*, was pinpointed because globular clusters crowd toward Sagittarius.

L5 Population I & II Stars

Astronomers call every element past helium a 'metal'. Population I stars like the Sun are metal-rich, young, and in the galactic disk — the ones that can have rocky planets; Population II are metal-poor first-generation stars in the halo and the globular clusters, formed before supernovae had enriched the gas.

TRAINER NOTE Week 2's band across the sky becomes a barred spiral seen from inside: Orion Arm for our address, Sgr A* at the centre — a black hole of last week's kind, four million Suns of it, which is why this session waits until now. Hubble's tuning fork: spiral, barred, elliptical, irregular. The Sagittarius Dwarf is colliding with us right now — one sentence, high quiz value. Nebula types by how they shine: emission, reflection, dark, planetary; open versus globular clusters alongside. Population I vs II is the reasoning bit — only Pop I, born from supernova-enriched gas, can have rocky planets; the quiz asks WHY, not which. The Messier table is pure memorisation and the quiz mines it: drill the celebrities cold — M1 Crab, M31 Andromeda, M42 Orion, M45 Pleiades.

12 Week 12 — Cosmology: the Big Bang and the expanding universe

CLASSROOM · 1 H 30 MIN

KIT Projector · Paper, pencils, protractor

L4 **Doppler & Redshift**

Motion stretches or squeezes light waves: approaching sources shift blue, receding ones red. Measured in spectral lines, it gives speeds of stars and galaxies to high precision.

L5 **The Expanding Universe**

Nearly every galaxy's light is redshifted, and farther means faster — space itself is stretching. Run the film backward and the universe had a beginning.

L5 **Big Bang & Cosmic History**

13.8 billion years from a hot dense start: relic microwave glow, primordial hydrogen and helium, and the growth of galaxies all tell one consistent story.

L5 **Dark Matter & Dark Energy**

Galaxies spin too fast for their visible mass, and the expansion is speeding up. Some 95% of the universe is stuff we can't yet see — astronomy's biggest open question.

L5 **The Cosmic Microwave Background**

The universe's baby photo: light from 380,000 years after the Big Bang, stretched to microwaves, arriving from every direction. Old TVs even picked up a whisper of it as static.

L5 **The First Elements**

For thousands of years the young universe was too hot for matter to condense out of energy — all of it fierce gamma radiation. As it cooled, that radiation formed neutrons, protons and electrons: about 91% hydrogen, 8% helium, under 1% lithium, with traces of deuterium and helium-3 — and essentially nothing heavier.

TRAINER NOTE WARNING THE TRAINER MUST CARRY: Study_Material_3 dates the universe at 12.5 billion years. That is WRONG — the accepted figure is 13.8 billion. Teach 13.8, tell the students the PDF errs, and tell them to write 13.8. The hour itself: Doppler first (the siren, then starlight), because Hubble's law is nothing but redshift plus distances. Expansion is space stretching, not an explosion INTO anything — the balloon surface, every raisin sees every other recede. Evidence in order: the redshifts, the CMB as the flash gone cold to 2.7 K, and the first three minutes leaving three-quarters hydrogen, one-quarter helium. Dark matter and dark energy honestly: names for measured ignorance, ~95% of the budget. Inflation gets one glossary-faithful sentence, no more.

13 Week 13 — India in space

CLASSROOM · 2 H

KIT Projector · Laptop · Paper, pencils, protractor

L2 **India in Space**

Chandrayaan-3 landed at the lunar south pole, Mangalyaan orbited Mars on a first attempt, Aditya-L1 stares at the Sun, AstroSat scans the X-ray sky — and Gaganyaan comes next.

L3 **Rockets & Launch Vehicles**

A rocket throws mass backwards to be pushed forwards, and sheds stages as they empty so it never hauls dead tanks to orbit. India's ladder runs SLV-3 (Rohini, 1980, the first Indian orbital launch), ASLV, the PSLV workhorse, GSLV with its indigenous cryogenic upper stage, LVM3 which sent up Chandrayaan-3, and the small-satellite SSLV.

L4 **Orbits & Artificial Satellites**

An orbit is perpetual falling that keeps missing the ground. Speed and altitude set the orbit: ISS circles in 92 minutes, geostationary satellites hover over one longitude.

L4 **Satellite Navigation & NavIC**

Each navigation satellite endlessly broadcasts the time by an atomic clock; your receiver hears several at once and finds the one spot on Earth where all those travel times agree — trilateration by clock, which is why the clocks must be extraordinary. India's NavIC (IRNSS-1A to 1I, now the NVS second generation) gives the region its own fix, independent of GPS.

L5 **Lagrange Points**

Five gravitational parking spots where a small body can hold station with two big ones. India's Aditya-L1 watches the Sun from L1; JWST observes from L2.

TRAINER NOTE The ISRO milestone table (1962 INCOSPAR through the Jan 2026 PSLV-C62 failure) is chronology — pure memorisation, and the densest question mine in the whole syllabus, so it went out as flashcards last week and gets tested today. The table is honest about failures (GSLV-D3, GSLV-F06, PSLV-C61) and quizmasters love failure questions precisely because students skip them. Physics before roll-call: an orbit is falling and missing, and a rocket is Week 5's third law with a fuel bill; then PSLV vs GSLV vs LVM3 by what each lifts. NavIC: seven satellites, regional, India's own. Aditya-L1's halo orbit is the Lagrange-point hook — L1 sees the Sun forever — and AstroSat gets its sentence as India's space telescope.

14 Week 14 — Telescopes, observatories, and the mock paper

INDOOR LAB · 2 H

KIT Projector · Laptop · Paper, pencils, protractor

L3 **Seeing & Transparency**
Seeing is atmospheric steadiness (how much stars boil); transparency is clarity (how faint you can go). A hazy still night can beat a crisp windy one for planets.

L3 **Choosing an Observatory Site**
Great observatories go high, dry and dark: altitude cuts the airmass and the water vapour that swallows infrared and submillimetre light, the Atacama is the driest place on Earth, and remoteness buys black skies. Mauna Kea sits at 4,190 m, Chile's TAO at 5,640 m, and India's own Indian Astronomical Observatory at Hanle in Ladakh at 4,500 m.

L3 **Refractors & Reflectors**
Refractors use a lens, reflectors a mirror, catadioptrics both. Each design trades cost, size, and image quality differently — there is no single 'best telescope'.

L4 **Radio Astronomy & the GMRT**
The sky broadcasts: pulsars tick, hydrogen clouds hum at 21 cm, galaxies roar. Near Pune stands the GMRT, one of the world's largest radio telescope arrays — and it's Indian.

L3 **Aperture & Light Grasp**
Aperture is the diameter of the main lens or mirror — it decides how faint you can see and how fine you can resolve. Aperture, not magnification, is a telescope's true power.

TRAINER NOTE Aperture is everything; magnification is nothing — if one idea survives this session, that one (resolution and light grasp both scale with the mirror, and the material's tables are sorted by it). Refractor vs reflector vs catadioptric by light path, then the material's mirror menagerie: segmented, single, honeycomb, liquid. Seeing explains why observatories flee upward, which is the whole logic of the two memorisation tables (Largest Telescopes, High Altitude Observatories) — India's Hanle IAO at 4,500 m sits in BOTH; flag it, it is the likeliest question in the section. Radio astronomy earns GMRT its mention; adaptive optics and interferometry get a sentence each, no more. Spend the back half on a timed mock paper drawn evenly from all fourteen weeks, marked in the room — the written round is a paper, so the course ends as one.

FULL KIT LIST

Laptop	sessions 13, 14
Projector	sessions 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14
Star chart / planisphere	sessions 2
Paper, pencils, protractor *(consumable)*	sessions 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14